

# Trading Desk v3

Project overview · idea, purpose and intended value

Build a small, auditable research desk that tests whether models, memory and carefully bounded AI assistance improve trading decisions after costs.

## The idea

Start by collecting market data, training and comparing algorithms, and testing them honestly on later, unseen periods. Then use a small decision-support team to explain model signals, challenge proposals and recall relevant past outcomes. A deterministic risk gate—not an AI prompt—controls what may proceed.

## Why build it?

- **Separate evidence from a good story.** Compare strategies with simple baselines, transaction costs and realistic timing before trusting them.
- **Keep continuity.** Connect a decision to its evidence, outcome and lesson so later decisions can use history without seeing the future.
- **Make decisions inspectable.** Preserve the reasoning, limits and approvals needed to understand why an action was allowed or rejected.

## The intended loop



Reviewed outcomes feed future research and eligible memory. Improvements are evaluated against a fixed baseline; they are not automatically adopted.

## What “learning” means here

### Numerical learning

Retrain models on dated evidence and compare new candidates with the existing strategy.

### Structured memory

Retrieve past lessons and propose policy improvements, with testing and human review before adoption.

**Scope:** daily, long/flat equities first; spot-crypto research stays separate. This is not an unrestricted AI trading bot, high-frequency platform or promise of returns. Real-money trading is optional and is not implemented in the starter.

# The architecture, simplified

Six domains, shared controls and a small starting stack

The architecture has **six responsibility layers (domains)** and two shared operating functions. These are clear boundaries inside a modular application—not a requirement to build six separate services.

## D1 Data and features

Collect, date and validate market observations; turn them into consistent numerical inputs.

## D2 Research and model lifecycle

Train and compare algorithms, test costs and holdouts, and preserve evidence for candidate models.

## D3 Decision support

Combine model signals and eligible lessons into explainable proposals. AI may assist with language, not execution authority.

## D4 Portfolio and risk

Value the book and check projected exposure, loss limits, available cash and approval validity.

## D5 Execution and accounting

Manage authorized orders, fills, fees and cash; reconcile the ledger with the broker. Broker integration is future work.

## D6 Memory and evaluation

Connect decisions to matured outcomes, retrieve relevant lessons and measure whether adaptation actually helps.

## Two functions span every layer

**Control plane:** human approvals, scheduling, environment selection, HALT, secrets, releases and recovery.

**Evidence plane:** decision history, data/model lineage, tests, metrics, alerts, audit export and backups.

## The most important boundary

**Models compute. Agents help explain. Risk enforces limits. Humans retain authority.** Agents cannot bypass the risk gate, approve their own actions or receive direct broker submission privileges. Unknown state should stop new risk, not invite a guess.

## Simple deployment first

One Python package and one writer on a laptop. Parquet stores research data; SQLite holds prototype state. LangGraph is optional for checkpoints and human pauses. Add databases, servers or cloud services only when measured needs justify them.

These layers describe the **target design**. The current prototype implements parts of them; full portfolio accounting, execution and operational safeguards still require development and certification.

# From idea to reliable operation

The phased roadmap and an honest view of readiness

Progress is controlled by **evidence gates, not calendar deadlines**. Research can produce a useful negative result; a model is not entitled to promotion because the tools work.

|   |                                   |                                                                                                                |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------------------|
| 0 | <b>Reproducible laptop</b>        | Install the locked environment and prove that offline tests and the CLI work.                                  |
| 1 | <b>Credible research</b>          | Validate data and model experiments; admit a candidate only with evidence—or promote nothing.                  |
| 2 | <b>Correct paper lifecycle</b>    | Build marked accounting, realistic paper execution, risk controls and replay-safe approvals.                   |
| 3 | <b>Measured memory</b>            | Test retrieval, policy and bounded adaptation against a fixed numerical baseline.                              |
| 4 | <b>Broker-paper certification</b> | Prove reconciliation and failure handling; consider a supervised tiny-live trial only after separate approval. |
| 5 | <b>Reliable operations</b>        | Monitor, back up, restore and manage releases; expand only through reviewed changes.                           |

Observation floors: 20 internal-paper equity sessions in Phase 2; 20 broker-paper sessions in Phase 4; 30 calendar days of operational evidence in Phase 5. These are not proof of profit or completion estimates.

## Available in the starter

- Local research, features and walk-forward screening.
- Rule-based reports, basic gates and a toy internal-paper book.
- Optional graph approvals, memory diagnostics and regression tests.

## Still required for operation

- Accurate marked/session accounting and realistic trade exits.
- Approved inference artifacts and time-correct memory.
- Broker reconciliation, scoped approvals, durable audit and recovery.

## What success looks like

Reproducible evidence, understandable decisions, enforced limits and dependable operations. If memory or AI does not improve the baseline, keep the simpler system. Staying paper-only—or retiring a weak strategy—is a valid outcome.

**Current status:** research and internal-paper prototype. No live broker adapter is provided. Tests and design documents do not establish profitable performance or production readiness.